

CSCI235 – Database Systems
2026 S2
Implementation Task 1
Due on 26 April 2026 by 9:00 pm SGT

Scope

The Implementation of Task 1 is related to the contents of topic on indexing.

This Implementation is due by Sunday, 26 April 2026, 9:00 pm Singapore time.
This task is worth 3% of the total assessment for the subject.

Only electronic submission through Moodle at:

<https://moodle.uowplatform.edu.au/>

is accepted. All email submission will be deleted and mark 0 ("zero") will be awarded.

The submission procedure is explained at the end of this specification.

Specification

Task 1 (1.5 marks)

1. Consider a relational schema $R(A, B, C, D, E)$ and the following set of functional dependencies valid in the schema:

(i) $\{A \rightarrow B, C \rightarrow D, E \rightarrow A\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

(ii) $\{A \rightarrow B, B \rightarrow CD, C \rightarrow E, E \rightarrow AC\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

(iii) $\{A \rightarrow B, DB \rightarrow C\}$

List all minimal keys valid in the schema. List all derivations of function dependencies that lead to the identification of minimal super keys. Note, a schema can have more than one minimal super key.

(0.5 mark)

Deliverable

A file Task1Solution.pdf with the solutions of the problems included in Task 1.

Submission of a file with a different name and/or different extension and/or different type scores no marks!

Task 2 (2.0 marks)

Read the following specification of sample database domain.

A hospital would like to create a medical database with the health records of its patients. Information recorded in the database supposed to include a medical insurance number that identifies each patient, first name, last name, date of birth, a list of admission dates to the hospital, list of illnesses cured during each admission and a list of children. Assume, that the first name, last name and date of birth uniquely identify each patient. Assume, that more than one illness can be cured during each admission. Also assume, that child is described by the first name and relation type to its parent, i.e. either son or daughter and all children of a patient have different names.

A database designer created the following relational schema.

MRECORD(insurance,fname,lname,dob,adate,illness,child,relation)

Determine what is the highest normal form the relational schema conforms to? Justify your choice.

Your next task is to decompose the relational schema into the smallest number of relational schemas, and each of the relational schema is normalized.

Deliverable

A file solution1.pdf with a list of the relational schemas obtained from the decomposition. Each relational schema must have attached a list of nontrivial functional and/or multivalued dependencies valid in the schema and a justification why the schema is normalized.

Submissions

This assignment is due by 9:00 pm (21:00 hours) Sunday, 26 April 2026, Singapore time.

Submit the files **IT1SolutionOutput.txt** through Moodle in the following way:

- 1) Access Moodle at **<http://moodle.uowplatform.edu.au/>**
- 2) To login use a Login link located in the right upper corner the Web page or in the middle of the bottom of the Web page
- 3) When successfully logged in, select a site CSCI235 (SP226) Database Systems
- 4) Scroll down to the section "Implementation Task 1 Submission link" and click at the link.
- 5) Click at a button Add Submission
- 6) Move the file that you want to submit into an area of submission. If you have more than one files to submit, you can drag and drop the remaining files here to add them. You can also use a link *Add...*
- 7) Click at the button "Save changes",
- 8) Click at the check box to confirm authorship of your submission and click at the button "Continue".
- 9) When you are satisfied, remember to click at a button Submit assignment.

A policy regarding late submissions is included in the subject outline. Only one submission per student is accepted.

Implementation Task 1 is an individual assignment, and it is expected that all its tasks will be solved individually without any cooperation with the other students. Plagiarism is treated seriously. Students involved will likely receive zero. If you have any doubts, questions, etc. please consult your lecturer or tutor during lab classes or over e-mail.